

REGRESSION

The term 'regression' literally means "stepping back towards the average". It was first used by a British Biometrician Sir Francis Galton.

The relationship between the independent and dependent variables may be expressed as a function. Such functional relationship between two variables is termed as regression.

In regression analysis independent variable is also known as regressor or predictor or explanatory variable while dependent variable is also known as regressed or explained variable.

When only two variables are involved the functional relationship is known as simple regression. If the relationship between two variables is a straight line, it is known as simple linear regression, otherwise it is called as simple non-linear regression.

Direct Method:

The regression equation of Y on X is given as

$$Y = a + bX$$

Where Y = dependent variable; X = independent variable and a = intercept

b = the regression coefficient (or slope) of the line. a and b are also called as Constants

the constants a and b can be estimated with by applying the 'least squares method'. This involves minimizing $\sum_i^n e^2 = \sum_i (y - a - bx)^2$. This gives

$$b_{yx} = b = \frac{Cov(XY)}{v(X)} = \frac{\sum XY - \frac{(\sum X)(\sum Y)}{n}}{\sum X^2 - \frac{(\sum X)^2}{n}}$$
$$\text{and } a = \bar{Y} - b\bar{X}$$

where b is called the estimate of regression coefficient of Y on X and it measures the change in Y for a unit change in X.

Similarly, the regression equation of X on Y is given as

$$X = a^1 + b^1Y$$

where X = dependent variable and Y = independent variable

$$b_{xy} = b^1 = \frac{Cov(XY)}{v(Y)} = \frac{\sum XY - \frac{(\sum X)(\sum Y)}{n}}{\sum Y^2 - \frac{(\sum Y)^2}{n}}$$

$$\text{and } a^1 = \bar{X} - b^1 \bar{Y}$$

where b^1 is known as the estimate of regression coefficient of X on Y and 'a' is intercept

Deviation Method:

The regression equation of Y on X is

$$(Y - \bar{Y}) = b(X - \bar{X})$$

$$\Rightarrow Y = \bar{Y} + b(X - \bar{X})$$

The regression equation of X on Y

$$(X - \bar{X}) = b^1(Y - \bar{Y})$$

$$\Rightarrow X = \bar{X} + b^1(Y - \bar{Y})$$

Properties of regression coefficient:

1. Correlation coefficient is the geometric mean of the two regression coefficients.
i.e $r = \pm \sqrt{b.b^1}$
2. If one of the regression coefficient is greater than unity, the other must be less than unity.
3. Arithmetic mean of the regression coefficients is greater than the correlation coefficient 'r'.
4. Regression coefficients are independent of the change of origin but not of scale.
5. Units of 'b' are same as that of the dependent variable.
6. Regression is only a one-way relationship between y (dependent variable) and x (independent variable).
7. The range of b is from $-\infty$ to $+\infty$. $-\infty$ for negative b and $+\infty$ for positive b.

Note:

1. Both the lines regression pass through the point $(\bar{X}, \bar{Y})$. In other words, the mean values $(\bar{X}, \bar{Y})$ can be obtained as the point of intersection of the two regression lines
2. If $r = 0$, the two variables are uncorrelated, the lines of regression become perpendicular to each other
3. If $r = \pm 1$, in this case the two lines of regression either coincide or they are parallel to each other
4. If the regression coefficients are positive, r is positive and if the regression coefficients are negative, r is negative

Distinguish between correlation and regression:

Correlation	Regression
1. Correlation is the relationship between two or more variables. Where the change in one variable affects a change in other variable 2. correlation coefficient measures extent of relationship between two variables 3. correlation is a two way relationship 4. correlation coefficient is independent of units of the variables. 5. correlation coefficient always lies between -1 and $+1$ 6. In correlation both the variables are random 7. correlation coefficient calculated by $r_{xy} = \frac{\text{cov}(x, y)}{\sqrt{V(x)V(y)}}$ 8. $r = \sqrt{bb^1}$ we can not predict	1. Regression is mathematical measure of the average relationship between two or more variables. Where one variable is dependent and other variable is independent 2. regression coefficient estimates the change in one variable for a unit change in other related variable 3. regression is a one way relationship 4. regression coefficient is expressed in the units of dependent variable 5. regression coefficient lies between $-\infty$ and $+\infty$ 6. In regression one variable will be in dependent and other will be dependent 7. the regression coefficient of y on x is $b = \frac{\sum xy - \frac{(\sum x)(\sum y)}{n}}{\sum x^2 - \frac{(\sum x)^2}{n}}$ the regression coefficient of x on y is $b^1 = \frac{\sum xy - \frac{(\sum x)(\sum y)}{n}}{\sum y^2 - \frac{(\sum y)^2}{n}}$ $b = r \sqrt{\frac{\sum (y - \bar{y})^2}{\sum (x - \bar{x})^2}}$; we can predict

Example: The following data give the yield per plant (gm) (Y) and days to flowering (X) of 11 pigeonpea plants. Fit regression equations of Y on X and X on Y. Also estimate Y when the flowering period is 149.

Solution:

The regression equation of Y on X is given by

$$Y = a + bX$$

Where Y = dependent variable; X = independent variable and a = intercept

b = the regression coefficient of Y on X

The constants a and b can be estimated with the help of 'least squares method'

$$\text{Where } b = \frac{\sum XY - \frac{(\sum X)(\sum Y)}{n}}{\sum X^2 - \frac{(\sum X)^2}{n}}; \quad a = \bar{Y} - b\bar{X}$$

Similarly, the regression equation of X on Y is given by

$$X = a^1 + b^1Y$$

Where X = dependent variable and Y = independent variable

$$\text{Where } b^1 = \frac{\sum XY - \frac{(\sum X)(\sum Y)}{n}}{\sum Y^2 - \frac{(\sum Y)^2}{n}}; \text{ and } a^1 = \bar{X} - b^1\bar{Y}$$

Y	X	Y ²	X ²	XY
24.72	91	611.08	8281	2249.52
20.25	76	410.06	5776	1539.00
38.56	98	1486.87	9604	3778.88
74.72	136	5583.08	18496	10161.92
72.75	128	5292.56	16384	9312.00
78.45	119	6154.40	14161	9335.55
69.8	142	4872.04	20164	9911.60
80.4	135	6464.16	18225	10854.00
160.2	147	25664.04	21609	23549.40
165.75	145	27473.06	21025	24033.75
77.56	122	6015.55	14884	9462.32
$\sum Y = 863.16$	$\sum X = 1339$	$\sum Y^2 = 90026.91$	$\sum X^2 = 168609$	$\sum XY = 114187.94$

$$b = \frac{114187.94 - \frac{(1339)(863.16)}{11}}{168609 - \frac{(1339)^2}{11}}$$

$$= \frac{9117.83}{5616.18}$$

$$= 1.62$$

$$\bar{X} = \frac{\sum X}{n} = \frac{1339}{11} = 121.73 \text{ and}$$

$$\bar{Y} = \frac{\sum Y}{n} = \frac{863.16}{11} = 78.47$$

$$\text{and } a = 78.47 - 1.62 * 121.73 = -118.74$$

The regression equation of Y on X is $Y = a + bX = Y = -118.74 + 1.62X$

To estimate Y when $X = 149$

$$\hat{Y} = -118.74 + 1.62 * 149 = 122.64$$

$$b^1 = \frac{114187.94 - \frac{(1339)(863.16)}{11}}{90026.91 - \frac{(863.16)^2}{11}} = \frac{9117.83}{22295.53} = 0.41$$

$$a^1 = \bar{X} - b^1 \bar{Y} = 121.73 - 0.41 * 78.47 = 89.56$$

The regression equation of X on Y is $X = 89.56 + 0.41Y$

Result:

1. The regression equation of Y on X is $Y = a + bX = Y = -118.74 + 1.62X$
2. The regression equation of X on Y is $X = 89.56 + 0.41Y$
3. Estimated Yield is 122.64